KUMARAGURU COLLEGE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science

DATA SCIENCE AND VISUALIZATION (24ADI204)

WEEK 1 REPORT

Team Formation & Data Hunting

Youth Smoking and Drug Consumption Analysis

DATA-VIZ HACK-A-BIT 2026–2027

Team 16

Team Member Name	Roll Number
Anireetha M	25BAD009

1. Week 1 Contribution

Researched feature engineering and PCA requirements.

2. Work Carried Out

During Week 1, my individual work focused on researching the requirements for feature engineering and Principal Component Analysis (PCA) for the Youth Smoking and Drug Consumption Analysis project. I reviewed the preprocessing steps that may be required before applying these techniques to the dataset.

3. Topics Researched

- **Feature engineering requirements –** Reviewed the need to identify numerical and categorical features and prepare them using suitable preprocessing methods.
- **Categorical encoding –** Studied Label Encoding for ordered drug-consumption categories and One-Hot Encoding for categorical values without a natural order.
- **Normalization and scaling –** Reviewed Standardization and Min-Max Scaling, and why numerical features may need scaling before further analysis.
- **Skewness –** Studied how to examine skewness in numerical features and the use of a Yeo-Johnson transformation to reduce skewness.
- **PCA requirements –** Researched PCA as a dimensionality-reduction technique, including the need to standardize numerical features and examine explained variance to decide how many principal components to retain.

4. Outcome of My Contribution

This research helped me understand the preprocessing requirements to consider before feature engineering and PCA. It also provided an overview of the techniques that could be explored in the later stages of the project. The work in this report is limited to researching these requirements; it does not claim that I implemented the feature engineering or PCA steps during Week 1.

5. Reference

Project notebook: DSV Team 16 – Youth Smoking and Drug Consumption Analysis.